

DIFFERENTIAL EQUATION:-

* We already know,

$$\frac{dy}{dx} \text{ (1st order)}, \frac{d^2y}{dx^2} \text{ (2nd order)}$$

$$\frac{d^3y}{dx^3} \text{ (3rd order) derivative.}$$

* Now Functional relation

$$F\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots, \frac{d^ny}{dx^n}\right) = 0 \text{ is}$$

called differential equation.

$$\text{Ex} * \frac{dy}{dx} = 2xy, * \frac{dy}{dx} = \sin x + \cos x$$

$$* \frac{d^2y}{dx^2} = 1 + \sqrt{\frac{dy}{dx}}.$$

* Order and Degree of D.E.

$$* \frac{d^3y}{dx^3} + 3\left(\frac{dy}{dx}\right)^2 + 7y = 0.$$

Here, highest order derivative is $\frac{d^3y}{dx^3}$

$\therefore$ order is 3

Now Degree :- Highest power of highest order derivative is 1

$$\left[\frac{d^3y}{dx^3} \text{ and its power is 1}\right]$$

$\therefore$ Degree : 1

$$* x \cdot \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^4 + xy = 0$$

$\Rightarrow$ Highest order derivative is $\frac{d^2y}{dx^2}$

$\therefore$ order is 2

Now $\frac{d^2y}{dx^2}$ has power 1

$\therefore$ Degree : 1

$$* x^2 \left(\frac{d^3y}{dx^3}\right)^2 + \left(\frac{dy}{dx}\right)^4 + y^2 = 0.$$

$\Rightarrow$ Highest order derivative is $\frac{d^3y}{dx^3}$

$\therefore$ order is 3

and $\left(\frac{d^3y}{dx^3}\right)$ its power is 2

$\therefore$ Degree is 2.

$$* \frac{dy}{dx} + \cos\left(\frac{dy}{dx}\right) = 0$$

$\Rightarrow$ Highest order derivative is $\frac{dy}{dx}$

$\therefore$ order is 1

But $\cos\left(\frac{dy}{dx}\right)$ cannot be expressed as Polynomial in $\frac{dy}{dx}$

$\therefore$ Degree is not define.

* Note: The terms containing $\log\left(\frac{dy}{dx}\right)$, $e^{\left(\frac{dy}{dx}\right)}$, $\sin\left(\frac{d^2y}{dx^2}\right)$, --- then Degree of D.E. is not define

$$* \frac{d^2y}{dx^2} = \sqrt[3]{1 + \left(\frac{dy}{dx}\right)^2}$$

First to make it radical free, cube both side

$$\Rightarrow \left(\frac{d^2y}{dx^2}\right)^3 = 1 + \left(\frac{dy}{dx}\right)^2.$$

Now Highest order derivative is

$\frac{d^2y}{dx^2}$ and its power is 3.

$\Rightarrow$ order : 2

degree : 3

* $x \cdot e^{\frac{dy}{dx}} + \frac{dy}{dx} + 3 = 0.$

Highest order derivative is $\frac{dy}{dx}$

$\Rightarrow$ order :- 1

Degree :- not define

($\because$ equation contain $e^{\frac{dy}{dx}}$).

$$\Rightarrow \left(\frac{d^2y}{dx^2}\right)^4 = \left(\frac{dy}{dx}\right)^9$$

$\Rightarrow$ order :- 2

Degree :- 4

* Linear Differential Equation.

* The order and degree of the differential equation $\left(\frac{d^3y}{dx^3}\right)^4 + \left(\frac{d^2y}{dx^2}\right)^3 = 0$ are _____.

$\Rightarrow$ order : 3, degree : 4

* The order of the differential equation $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = xy$ is _____.

$\Rightarrow$ order : 2

* The degree of the differential equation $x^2 \cdot \frac{dy}{dx} + \sin\left(\frac{d^2y}{dx^2}\right) = 0$ is _____.

$\Rightarrow$ degree is not define.

* The order and degree of differential equation $\frac{d^2y}{dx^2} = 3 \sin 3x$ are _____

$\Rightarrow$ order :- 2

degree :- 1

* The order and degree of differential equation $\sqrt[3]{\left(\frac{d^2y}{dx^2}\right)^2} = \sqrt{\left(\frac{dy}{dx}\right)^3}$ are _____.

$\Rightarrow$ First we make it free from radical

$$\left(\frac{d^2y}{dx^2}\right)^{2/3} = \left(\frac{dy}{dx}\right)^{3/2}$$

Taking '6' power on both side.

$$\left(\frac{d^2y}{dx^2}\right)^{2/3 \times 6} = \left(\frac{dy}{dx}\right)^{3/2 \times 6}$$

* L.D.E whose form is

$$\frac{dy}{dx} + Py = Q.$$

$\Rightarrow$ Integrating Factor (I.F.) = $e^{\int P dx}$

$$\text{Solution : } y \cdot (I.F.) = \int [Q \cdot (I.F.)] dx + C.$$

* L.D.E. whose form is

$$\frac{dx}{dy} + Px = Q$$

$$\Rightarrow I.F. = e^{\int P dy}$$

$$\text{Solution : } x \cdot (I.F.) = \int [Q \cdot (I.F.)] dy + C$$

* An integrating factor of the equation

$$\frac{dy}{dx} + y = 3x \text{ is } \underline{\hspace{2cm}}.$$

$$\Rightarrow \text{Here } P = 1, \Rightarrow I.F. = e^{\int P dx} = e^{\int 1 dx}$$

$$I.F. = e^x$$

* An integrating factor of the equation

$$\frac{dy}{dx} + \frac{2y}{x} = e^x \text{ is } \underline{\hspace{2cm}}.$$

$$\Rightarrow \text{Here } P = \frac{2}{x} \Rightarrow I.F. = e^{\int P dx} = e^{\int \frac{2}{x} dx} = e^{2 \log x} = e^{\log x^2}$$

$$I.F. = x^2$$

* An integrating factor of the equation

$$\frac{dy}{dx} = y \tan x + e^x \text{ is } \underline{\hspace{2cm}}$$

$$\Rightarrow \text{Here } P = -\tan x \Rightarrow I.F. = e^{\int P dx} = e^{\int -\tan x dx} = e^{-\log \sec x} = e^{\log \sec x} = \sec x = \frac{1}{\cos x}$$

* Solution of Differential Equation:

* Verify that $y = \frac{a}{x} + b$ is a general solution of the differential equation

$$\frac{d^2 y}{dx^2} + \frac{2}{x} \frac{dy}{dx} = 0.$$

$$\Rightarrow y = \frac{a}{x} + b$$

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx} \left(\frac{a}{x} + b \right) \\ = -\frac{a}{x^2} + 0 = -\frac{a}{x^2}.$$

$$\Rightarrow \frac{d^2 y}{dx^2} = \frac{d}{dx} \left(-\frac{a}{x^2} \right) = -a \frac{d}{dx} \left(\frac{1}{x^2} \right) \\ = -a \cdot \left(-\frac{2}{x^3} \right) \\ \frac{d^2 y}{dx^2} = \frac{2a}{x^3}$$

$$\text{Now, } \frac{d^2 y}{dx^2} + \frac{2}{x} \frac{dy}{dx} \\ = \frac{2a}{x^3} + \frac{2}{x} \left(-\frac{a}{x^2} \right) \\ = \frac{2a}{x^3} - \frac{2a}{x^3} = 0.$$

* Find the differential equation for $y = a \sin(x+b)$, where a and b are const.

$$\Rightarrow y = a \sin(x+b)$$

$$\frac{dy}{dx} = \frac{d}{dx} [a \sin(x+b)] \\ = a \cos(x+b) \cdot \frac{d}{dx} (x+b)$$

$$\frac{dy}{dx} = a \cos(x+b)$$

$$\Rightarrow \frac{d^2 y}{dx^2} = \frac{d}{dx} [a \cos(x+b)] \\ = a (-\sin(x+b)) \cdot \frac{d}{dx} (x+b)$$

$$\Rightarrow \frac{d^2 y}{dx^2} = -a \sin(x+b)$$

$$\Rightarrow \frac{d^2 y}{dx^2} = -y$$

$$\Rightarrow \frac{d^2 y}{dx^2} + y = 0 \\ \text{which is required equation.}$$

* Separable Variable

* If the D.E. can be put in the form $g(x)dx = g(y)dy$

[i.e. In given D.E we can write variable 'x' and its derivative on one side and variable 'y' and its derivative on another side]

$\Rightarrow$ Then take integration on both side

The solution is given by

$$\int g(x) dx = \int g(y) dy + C$$

* Solve the differential equation $x dy + y dx = 0$.

$$\Rightarrow x dy + y dx = 0$$

$$\Rightarrow x dy = -y dx$$

$$\Rightarrow + \frac{dy}{y} = - \frac{dx}{x}$$

$$\Rightarrow \int \frac{dy}{y} = - \int \frac{dx}{x}$$

$$\Rightarrow \log y = -\log x + \log C$$

$$\Rightarrow \log x + \log y = \log C$$

$$\Rightarrow \log(x \cdot y) = \log C$$

$$\Rightarrow x \cdot y = C$$

* Solve the differential equation

$$y(1+x)dx + x(1-y)dy = 0$$

$$\Rightarrow y(1+x)dx + x(1-y)dy = 0$$

$$\Rightarrow y(1+x)dx = -x(1-y)dy$$

$$\Rightarrow \frac{1+x}{x} dx = - \left(\frac{1-y}{y} \right) dy$$

$$\Rightarrow \frac{1+x}{x} dx + \left(\frac{1-y}{y} \right) dy = 0$$

$$\Rightarrow \left[\frac{1}{x} + 1 \right] dx + \left[\frac{1}{y} - 1 \right] dy = 0$$

$$\Rightarrow \int \left(\frac{1}{x} + 1\right) dx + \int \left(\frac{1}{y} - 1\right) dy = 0$$

$$\Rightarrow \log x + x + \log y - y = C$$

$$\Rightarrow \log x + \log y + x - y = C$$

$$\Rightarrow \log(x \cdot y) + x - y = C$$

* Solve the differential equation

$$x(1+y^2) dx = y(1+x^2) dy$$

$$\Rightarrow x(1+y^2) dx = y(1+x^2) dy$$

$$\Rightarrow \frac{x}{1+x^2} dx = \frac{y}{1+y^2} dy$$

$$\Rightarrow \frac{2x}{1+x^2} dx = \frac{2y}{1+y^2} dy$$

$$\Rightarrow \int \frac{2x}{1+x^2} dx = \int \frac{2y}{1+y^2} dy$$

$$\begin{aligned} & \# \int \frac{f'(x)}{f(x)} dx \\ & = \log f(x) + C \end{aligned}$$

$$\Rightarrow \log(1+x^2) = \log(1+y^2) + \log C$$

$$\Rightarrow \log(1+x^2) = \log(1+y^2) \cdot C$$

$$\Rightarrow (1+x^2) = (1+y^2) \cdot C$$

$$\Rightarrow \frac{1+x^2}{1+y^2} = C.$$

* Solve the differential equation

$$\frac{dy}{dx} = 1 + x + y + xy$$

$$\Rightarrow \frac{dy}{dx} = 1 + x + y + xy$$

$$\Rightarrow \frac{dy}{dx} = (1+x) + y(1+x)$$

$$\Rightarrow \frac{dy}{dx} = (1+x) \cdot (1+y)$$

$$\Rightarrow \frac{dy}{1+y} = (1+x) dx$$

$$\Rightarrow \int \frac{dy}{1+y} = \int (1+x) dx$$

$$\Rightarrow \log(1+y) = x + \frac{x^2}{2} + C.$$

* Solve the differential equation

$$x \cdot \frac{dy}{dx} + \cot y = 0.$$

$$\Rightarrow x \cdot \frac{dy}{dx} + \cot y = 0$$

$$\Rightarrow x \cdot \frac{dy}{dx} = -\cot y$$

$$\Rightarrow \frac{dy}{\cot y} = -\frac{dx}{x}$$

$$\Rightarrow \frac{dy}{\cot y} + \frac{dx}{x} = 0$$

$$\Rightarrow \int \frac{dy}{\cot y} + \int \frac{dx}{x} = 0$$

$$\Rightarrow \int \frac{1}{\cot y} dy + \int \frac{1}{x} dx = 0$$

$$\Rightarrow \int \tan y dy + \int \frac{1}{x} dx = 0$$

$$\Rightarrow \log(\sec y) + \log x = \log C$$

$$\Rightarrow \log(\sec y \cdot x) = \log C$$

$$\Rightarrow \sec y \cdot x = C.$$

* Solve the differential equation

$$\sec^2 x \cdot \tan y dx + \sec^2 y \cdot \tan x dy = 0.$$

$$\Rightarrow \sec^2 x \cdot \tan y dx = -\sec^2 y \cdot \tan x dy$$

$$\Rightarrow \frac{\sec^2 x}{\tan x} dx = -\frac{\sec^2 y}{\tan y} dy$$

$$\Rightarrow \frac{\sec^2 x}{\tan x} dx + \frac{\sec^2 y}{\tan y} dy = 0.$$

$$\Rightarrow \int \frac{\sec^2 x}{\tan x} dx + \int \frac{\sec^2 y}{\tan y} dy = 0.$$

$$\Rightarrow \log(\tan x) + \log(\tan y) = \log C$$

$$\Rightarrow \log[\tan x \cdot \tan y] = \log C$$

$$\Rightarrow \tan x \cdot \tan y = C.$$

$$\begin{aligned} & \# \int \frac{f'(x)}{f(x)} dx \\ & = \log f(x) + C \end{aligned}$$

* Solve the differential equation

$$\tan y dx + \tan x \cdot \sec^2 y dy = 0.$$

$$\Rightarrow \frac{dx}{\tan x} + \frac{\sec^2 y}{\tan y} dy = 0.$$

$$\Rightarrow \int \frac{1}{\tan x} dx + \int \frac{\sec^2 y}{\tan y} dy = 0.$$

$$\Rightarrow \int \cot x dx + \int \frac{\sec^2 y}{\tan y} dy = 0.$$

$$\Rightarrow \log(\sin x) + \log(\tan y) = \log C$$

$$\Rightarrow \log[\sin x \cdot \tan y] = \log C$$

$$\Rightarrow \sin x \cdot \tan y = C.$$

* Linear Differential Equation:-

* Method of solving a. L. D. E.

Let, $\frac{dy}{dx} + Py = Q$ be a given L.D.E.

$\Rightarrow$ Find, P, Q.

Now Find I.F. = $e^{\int P dx}$.

Then solution is

$$y \cdot (I.F.) = \int Q \cdot (I.F.) dx + C.$$

* Solve the differential equation

$$\frac{dy}{dx} + \frac{y}{x} = x^2 \Rightarrow \frac{dy}{dx} + Py = Q$$

$$\Rightarrow P = \frac{1}{x}, Q = x^2$$

$$I.F. = e^{\int P dx} = e^{\int \frac{1}{x} dx} = e^{\log_e x} = x$$

$$\Rightarrow I.F. = x.$$

The solution is

$$y \cdot (I.F.) = \int Q \cdot (I.F.) dx + C$$

$$\Rightarrow y \cdot x = \int x^2 \cdot x dx + C$$

$$\Rightarrow xy = \int x^3 dx + C$$

$$\Rightarrow xy = \frac{x^4}{4} + C.$$

* Solve the differential equation

$$\frac{dy}{dx} + \frac{y}{x} = e^x$$

$$\Rightarrow P = \frac{1}{x}, Q = e^x$$

$$I.F. = e^{\int P dx} = e^{\int \frac{1}{x} dx} = e^{\log_e x} = x$$

$$\Rightarrow I.F. = x.$$

The solution is

$$y \cdot (I.F.) = \int Q \cdot (I.F.) dx + C$$

$$\Rightarrow y \cdot x = \int e^x \cdot x dx + C$$

$$\Rightarrow y \cdot x = x \cdot e^x - \int (1 \cdot e^x) dx + C$$

$$\boxed{\int u \cdot v dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx}$$

$$\Rightarrow y \cdot x = x \cdot e^x - \int e^x dx + C$$

$$\Rightarrow y \cdot x = x \cdot e^x - e^x + C$$

* Solve the differential equation.

$$(1+x^2) \frac{dy}{dx} + 2xy = \cos x$$

$$\Rightarrow \frac{dy}{dx} + \frac{2x}{1+x^2} y = \frac{\cos x}{1+x^2}$$

$$\Rightarrow P = \frac{2x}{1+x^2}, Q = \frac{\cos x}{1+x^2}$$

$$I.F. = e^{\int P dx} = e^{\int \frac{2x}{1+x^2} dx} = e^{\log_e (1+x^2)} = 1+x^2$$

$$\boxed{\int \frac{f(x)}{g(x)} dx = \log f(x)}$$

$$\Rightarrow I.F. = 1+x^2$$

The solution is

$$y \cdot (I.F.) = \int Q \cdot (I.F.) dx + C$$

$$\Rightarrow y \cdot (1+x^2) = \int \frac{\cos x}{1+x^2} \cdot (1+x^2) dx + C$$

$$\Rightarrow y \cdot (1+x^2) = \int \cos x dx + C$$

$$\Rightarrow y \cdot (1+x^2) = \sin x + C.$$

* Solve the differential equation

$$\frac{dy}{dx} + y \cot x = \cos x$$

$$\Rightarrow P = \cot x, Q = \cos x.$$

$$I.F. = e^{\int P dx} = e^{\int \cot x dx} = e^{\log_e \sin x} = \sin x$$

$$\Rightarrow I.F. = \sin x$$

The solution is

$$y \cdot (I.F.) = \int Q \cdot (I.F.) dx + C$$

$$\Rightarrow y \cdot \sin x = \int \cos x \cdot \sin x dx + C$$

$$\Rightarrow y \cdot \sin x = \frac{1}{2} \int 2 \sin x \cdot \cos x dx + C$$

$$\Rightarrow y \cdot \sin x = \frac{1}{2} \int \sin 2x dx + C$$

$$\Rightarrow y \cdot \sin x = \frac{1}{2} \cdot \left(-\frac{\cos 2x}{2} \right) + C$$

$$\Rightarrow y \cdot \sin x = -\frac{\cos 2x}{4} + C$$

* Solve the differential equation

$$\frac{dy}{dx} + y \cdot \tan x = \cos x$$

$$\Rightarrow P = \tan x, \quad Q = \cos x$$

$$\text{I.F.} = e^{\int P dx} = e^{\int \tan x dx} = e^{\log_e \sec x} = \sec x$$

$$\Rightarrow \text{I.F.} = \sec x$$

The solution is

$$y \cdot (\text{I.F.}) = \int Q \cdot (\text{I.F.}) dx + C$$

$$\Rightarrow y \cdot \sec x = \int \cos x \cdot \sec x dx + C$$

$$\Rightarrow y \cdot \sec x = \int 1 dx + C$$

$$\Rightarrow y \cdot \sec x = x + C$$

* Solve the differential equation

$$\cos^2 x \frac{dy}{dx} + y = \tan x$$

$$\Rightarrow \frac{dy}{dx} + \frac{1}{\cos^2 x} \cdot y = \frac{\tan x}{\cos^2 x}$$

$$\Rightarrow \frac{dy}{dx} + \sec^2 x \cdot y = \frac{\tan x}{\cos^2 x}$$

$$\Rightarrow P = \sec^2 x, \quad Q = \frac{\tan x}{\cos^2 x}$$

$$\text{I.F.} = e^{\int P dx} = e^{\int \sec^2 x dx} = e^{\tan x}$$

$$\Rightarrow \text{I.F.} = e^{\tan x}$$

The solution is

$$y \cdot (\text{I.F.}) = \int Q \cdot (\text{I.F.}) dx + C$$

$$\Rightarrow y \cdot e^{\tan x} = \int \frac{\tan x}{\cos^2 x} \cdot e^{\tan x} dx + C$$

$$\Rightarrow y \cdot e^{\tan x} = \int \tan x \cdot \sec^2 x \cdot e^{\tan x} dx + C \quad \text{--- (1)}$$

Now in R.H.S. take $u = \tan x$
 $\Rightarrow du = \sec^2 x dx$

$$\Rightarrow \int \tan x \cdot \sec^2 x \cdot e^{\tan x} dx = \int u \cdot e^u du$$

$$= u \cdot e^u - \int (1 \cdot e^u) du + C$$

$$= u \cdot e^u - \int e^u du + C$$

$$= u \cdot e^u - e^u + C$$

$$= e^u (u - 1) + C$$

$$= e^{\tan x} (\tan x - 1) + C$$

Now from (1)

$$\Rightarrow y \cdot e^{\tan x} = e^{\tan x} (\tan x - 1) + C$$

* Solve the differential equation

$$x \log x \frac{dy}{dx} + y = \log x^2$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{x \cdot \log x} = \frac{\log x^2}{x \cdot \log x}$$

$$\Rightarrow \frac{dy}{dx} + \frac{1}{x \cdot \log x} y = \frac{2 \log x}{x \cdot \log x} \quad \leftarrow \log x^2 = 2 \log x$$

$$\Rightarrow \frac{dy}{dx} + \frac{1}{x \cdot \log x} y = \frac{2}{x}$$

$$\Rightarrow P = \frac{1}{x \cdot \log x}, \quad Q = \frac{2}{x}$$

$$\text{I.F.} = e^{\int P dx} = e^{\int \frac{1}{x \cdot \log x} dx} = e^{\int \frac{1/x}{\log x} dx}$$

$$= e^{\log(\log x)} \quad \leftarrow \int \frac{f'(x)}{f(x)} dx = \log f(x)$$

$$\Rightarrow \text{I.F.} = \log x$$

The solution is

$$y \cdot (\text{I.F.}) = \int Q \cdot (\text{I.F.}) dx + C$$

$$\Rightarrow y \cdot \log x = \int \frac{2}{x} \cdot \log x dx + C$$

$$\Rightarrow y \cdot \log x = 2 \int \frac{\log x}{x} dx + C \quad \text{--- (1)}$$

In R.H.S take $u = \log x$
 $\Rightarrow du = \frac{1}{x} dx$

$$\Rightarrow \int \frac{\log x}{x} dx = \int u du$$

$$= \frac{u^2}{2} + C$$

$$= \frac{(\log x)^2}{2} + C$$

Now from (1)

$$\Rightarrow y \cdot \log x = 2 \frac{(\log x)^2}{2} + C$$

$$\Rightarrow y \cdot \log x = (\log x)^2 + C$$